

Climate Forcing and Timescales

1. Forcing, non-equilibrium, and return to equilibrium

- Using the information from this week's reading (*Dessler*, Chapter 6) and lecture, give a precise definition of **radiative forcing** (RF). Recall that it is a difference of two fluxes. What are its units?
- Explain the *meaning* of radiative forcing in the context of climate (radiative) equilibrium. Make reference to our prior work using energy flow/flux diagrams, and in particular the relationship of forcing to *equilibrium*. Explain the sequence of energy flux diagrams in *Dessler*'s Figure 6.1 (reproduced here in Figure 1). What is the value (and units) of the radiative forcing imposed by the sudden atmosphere?
- Repeat the calculation example done by *Dessler* in Section 6.3.1. Assume that the solar flux initially has its standard value¹ of 238 W/m^2 . Then, because of an orbital event, the Sun's flux increases by 1 W/m^2 , i.e., there is a radiative forcing of $+1 \text{ W/m}^2$. If the Earth's original equilibrium temperature (assuming no atmosphere) was 254.55 K , what will be the new temperature after it returns to equilibrium?
- The radiative forcing induced by all of the human carbon dioxide production since 1750 (changing the atmospheric concentration from 280 ppm to 425 ppm) is about $+2.2 \text{ W/m}^2$. The **climate sensitivity** — the forcing induced by a doubling of atmospheric carbon dioxide — is about $+4.0 \text{ W/m}^2$. Using your answer to part (c), which gave an estimate of the temperature increase per 1 W/m^2 of forcing, what are the corresponding temperature increase² predicted by these forcings for a no-atmosphere Earth?

2. Timescales of atmospheric carbon dioxide and temperature increase

- The change in atmospheric carbon dioxide over the past 70 years (the “Keeling Curve”) is shown in Figure 2. Let's try to characterize that data by plotting it ourselves.
 - First make a table with columns $[t(\text{yr}), C(\text{ppm}), \log_{10}(C)]$. Then choose about 5 points from the Keeling Curve data and record the values in your table, also calculating the logarithm of the concentration, C , value in your calculator.
 - Take a regular sheet of graph paper and plot $\log_{10}(C)$ vs time. Be sure to choose your axis ranges so that you can fill most of the paper.
 - Draw a best fit line to your plotted data and find its slope.

When a quantity is increasing exponentially, it can be seen as a straight line on a log vs linear plot. Does this data show evidence of exponential growth?

- For an exponentially growing quantity, we can assign a constant “doubling time” to the growth rate. This value is found as

$$t_{\text{double}} = \frac{\log_{10}(2)}{\text{slope}}$$

¹Remember that this value is $F_{\text{sun}} = \frac{S}{4}(1 - \alpha)$, where $S = 1360 \text{ J/s/m}^2$ is the Solar Constant, and $\alpha = 0.3$ is the albedo.

²Feedbacks in the climate system (e.g., ice melt $\rightarrow$ decreasing albedo $\rightarrow$ increasing F_{sun} , see *Dessler* Section 6.3.2) amplify the climate sensitivity. Current models incorporating all known effects estimate climate sensitivity to be about $3 \text{ K} = 3^\circ\text{C}$, with a likely range ($\sim 70\%$ probability of being in) 2.6 to 4°C .

where “slope” is the slope of the log-linear plot. Using this equation and your result from part (a), find the doubling time for atmospheric carbon dioxide.

- (c) Considering a longer-time plot of carbon dioxide in the atmosphere (Figure 3) and assuming that the exponential growth rate held over this whole period. When do you expect we will double the value of carbon dioxide from 1750?

3. (**Not used in F2025**) Consider the graphs in Figure 4. The top graph shows the effect of an annual 1% increase in CO₂ up to doubling. The bottom shows that the temperature is, model-predicted, to be proportional to the cumulative amount of emissions.

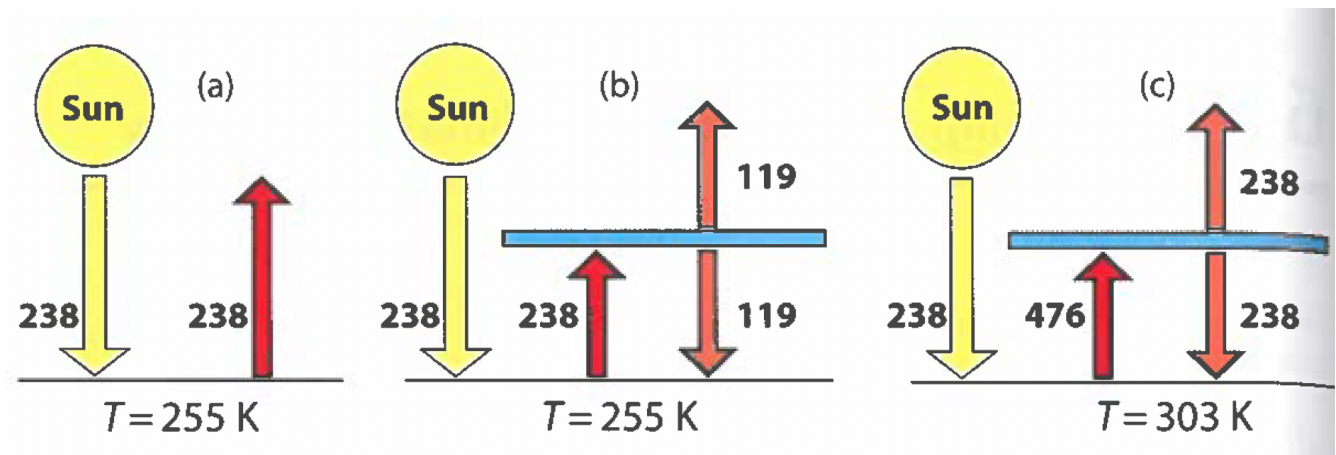


Figure 1: Figure 6.1 from *Dessler*: (a) An energy flux diagram for an Earth with no atmosphere (in equilibrium). (b) The immediate imposition of an atmosphere which is able to capture the Earth surface's IR radiation and re-radiate it. (c) A return to equilibrium.

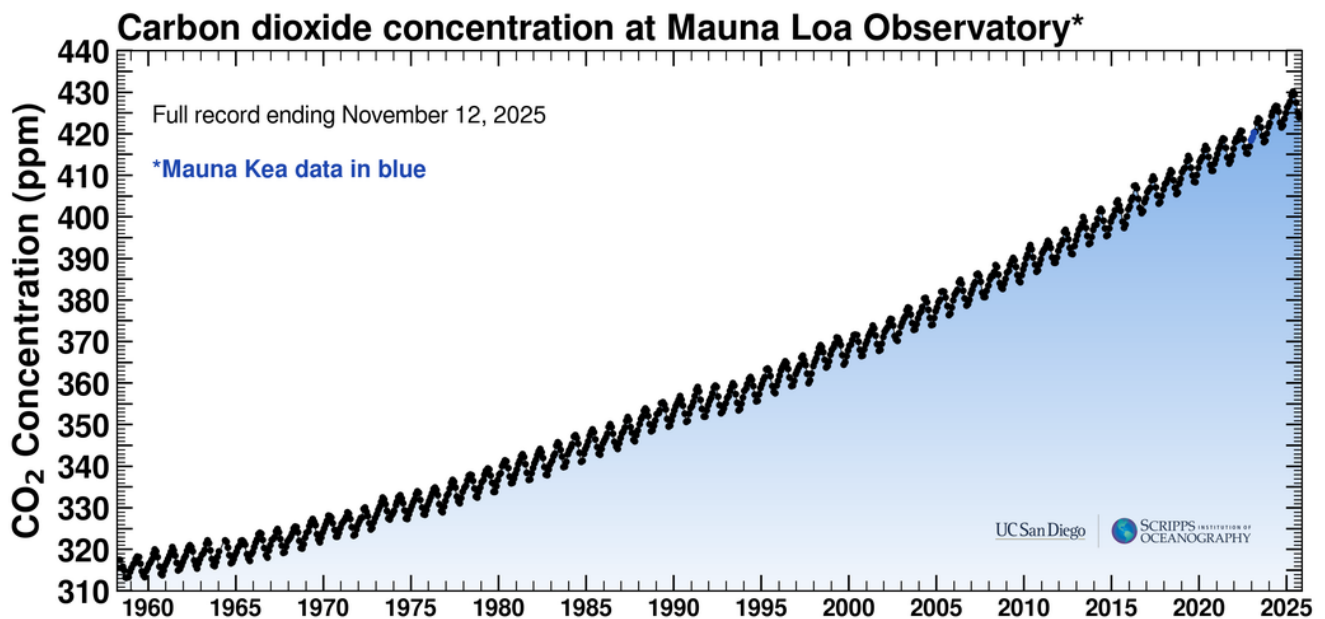


Figure 2: The so-called **Keeling Curve** (named after Charles Keeling, who began atmospheric CO₂ measurements at Mauna Loa Observatory in Hawaii, in 1956) showing atmospheric carbon dioxide as a function of time.

Global carbon dioxide emissions and atmospheric carbon dioxide (1751-2024)

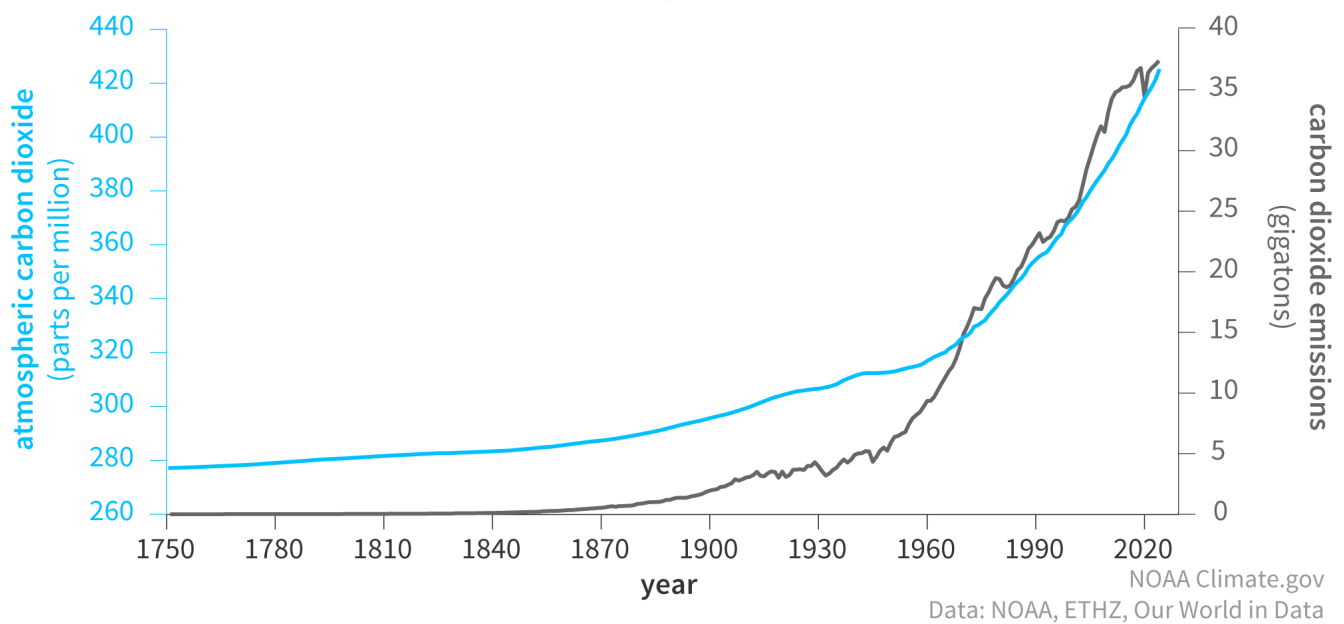


Figure 3: Atmospheric carbon dioxide and emissions since the Industrial Revolution.

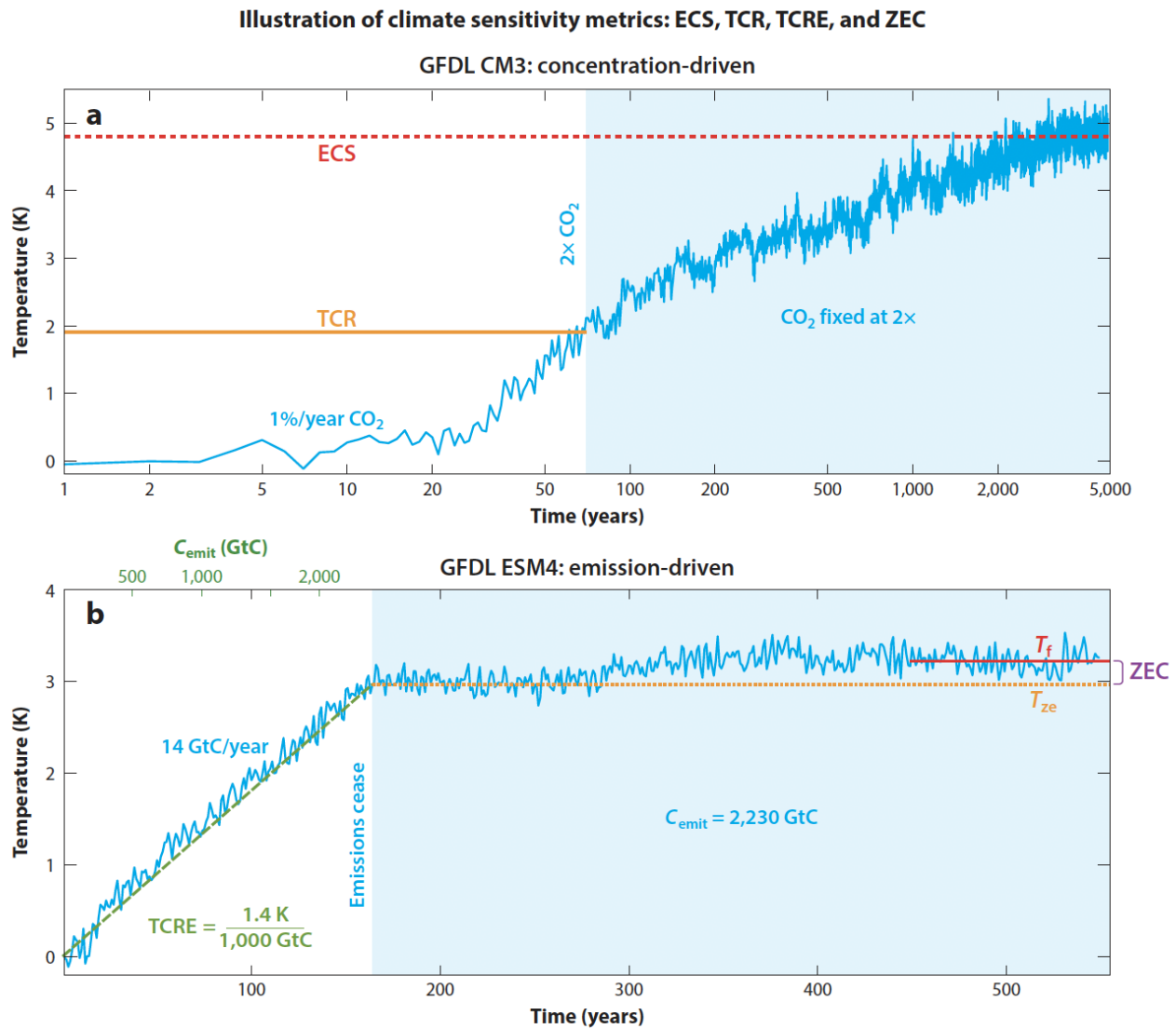


Figure 1

(a) Global mean surface temperature T_s versus time for a multi-millennial atmosphere-ocean GCM simulation using GFDL CM3 (Donner et al. 2011, Paynter et al. 2018). CO₂ concentrations increase at 1%/year until doubling at year 70 and are held fixed thereafter. TCR is the warming at year 70, and ECS is the equilibrium warming after several millennia. Note the log time axis.

(b) T_s versus time for an Earth system simulation with an active carbon cycle using GFDL ESM4. Constant emissions of roughly 14 GtC/year are prescribed until roughly 3°C of warming, at which point emissions cease. The TCRC is the slope of T_s versus cumulative emissions C_{emit} (green axis, upper left) while emissions are nonzero, and ZEC is the difference between the final warming T_f and the warming when emissions are zeroed T_{ze} . Abbreviations: ECS, equilibrium climate sensitivity; GFDL, Geophysical Fluid Dynamics Laboratory; GtC, gigatons carbon; TCR, transient climate response; TCRC, transient climate response to cumulative emissions; ZEC, zero emissions commitment.

Figure 4: Two pathways to the future.